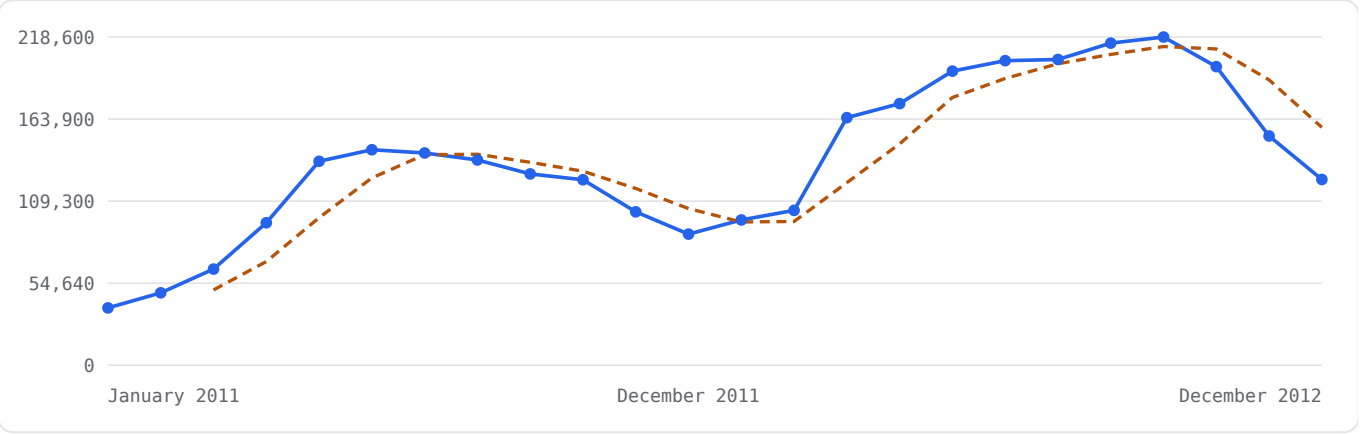
# Two years of Washington bike sharing

Created 3 November 2026 with Data Boss · Data: day.csv

Capital Bikeshare, Washington D.C., 1 January 2011 – 31 December 2012: 731 days of rentals with weather (UCI Bike Sharing dataset, CC BY 4.0).

### Trend: sum of cnt per month

- March 2012 was 60% above the previous month (164,875), the biggest rise.
- November 2012 was 23% below the previous month (152,664), the biggest drop.
- Highest: 218,573 in September 2012. Lowest: 38,189 in January 2011.
- The latest complete month, December 2012, is 19% below the previous month.
- Overall the trend is rising: about 5,453 more per month (4.0% of the average).




### Welch's t-test: Does casual differ between groups of day_type?

casual differs between working day and weekend / holiday (p < 0.001): weekend / holiday averages 765 more than working day (95% CI 646 to 883). The difference is large.

| < 0.001 | -1.30 |
|---|---|
| p-value | Cohen's d (large) |

| -882.8 … -646.3 | 731 |
|---|---|
| 95% CI · Difference in mean casual (working day − weekend / holiday) | Rows used |

Why this test: Two groups with a roughly normal shape; spreads differ, so Welch's version.

- This data was observed, not from an experiment: a difference or relationship does not show what causes it.

### One-way ANOVA: Does cnt differ between groups of weather?

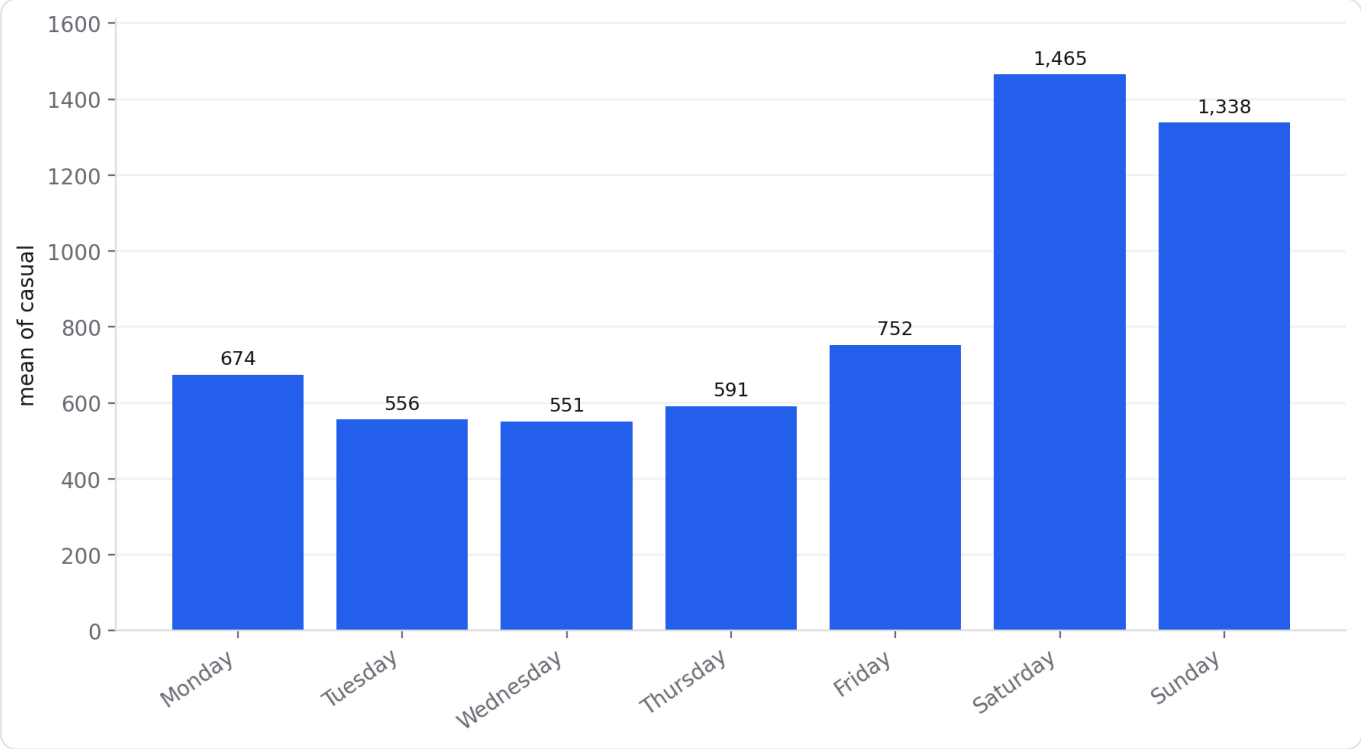
> cnt is not the same across all 3 groups of weather (p < 0.001); the average ranges from 1,803 (light rain / snow) to 4,877 (clear). The effect is medium.

| < 0.001 | 0.10 | 731 |
|---|---|---|
| p-value | η² (eta squared) (medium) | Rows used |

Why this test: 3 groups with a roughly normal shape and similar spread.

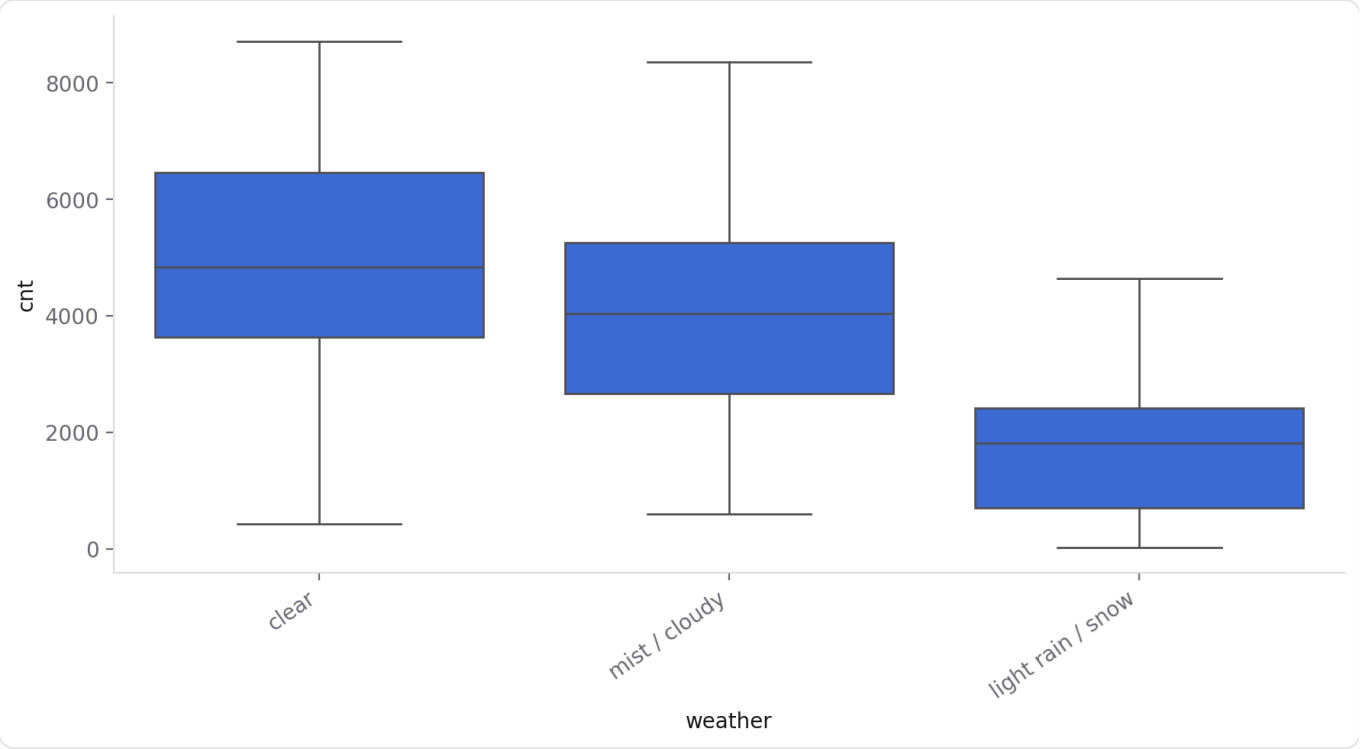
- The overall test only says some groups differ; the pairwise table shows which (p-values adjusted for making several comparisons).
- Group sizes are very unequal; the smaller groups have much wider uncertainty.
- This data was observed, not from an experiment: a difference or relationship does not show what causes it.

### Average casual riders by weekday



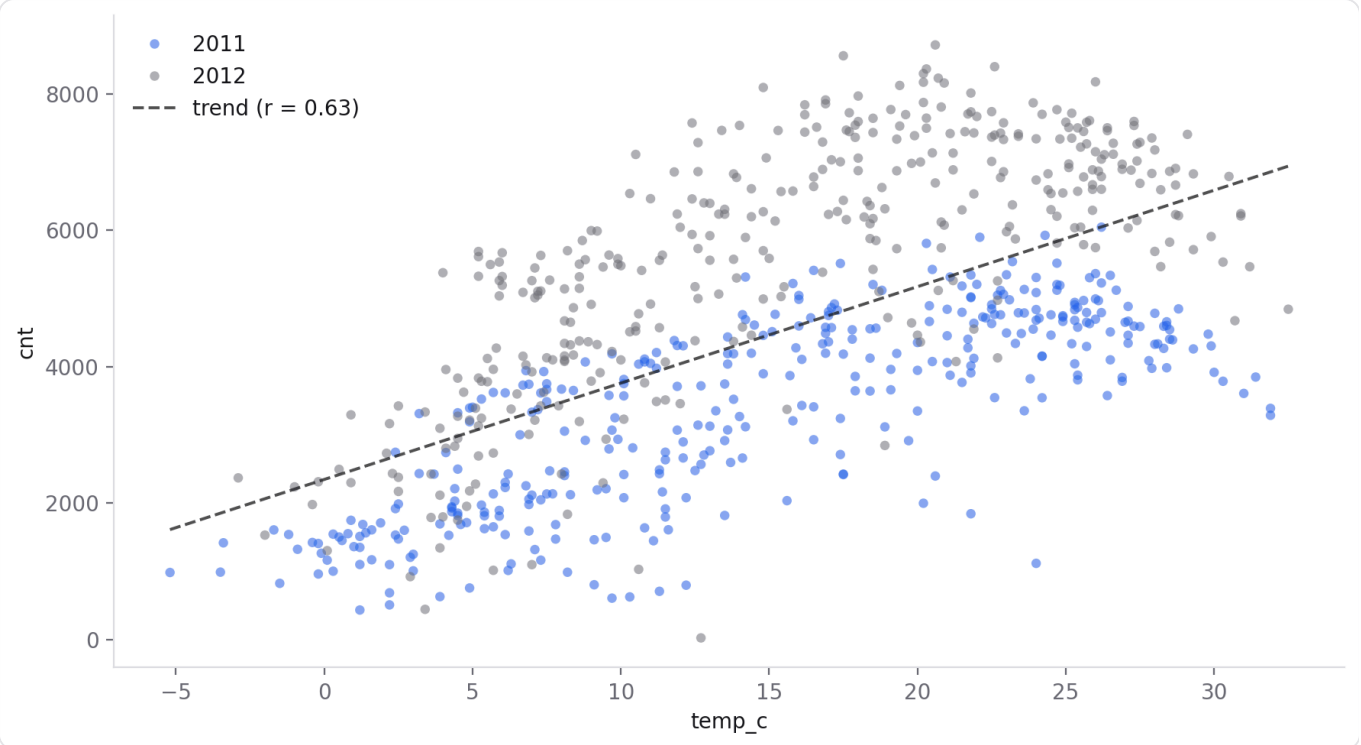

- Highest: Saturday (1,465). Lowest: Wednesday (551).
- The highest group is 2.7× the lowest.

### Daily rentals by weather




- Highest median: clear (4,844). Lowest: light rain / snow (1,817).
- Overall median 4,548; middle 50% between 3,152 and 5,956.
- No points outside the whiskers.

### Daily rentals against temperature (°C)




- Pearson r = 0.63: strong positive relationship (731 points).
- Trend line: each +1 in temp_c goes with about +141 in cnt.

## Forecast: sum of cnt per day (exponential smoothing with level and seasonality)

| Period | Forecast | Low (80%) | High (80%) |
|---|---|---|---|
| 01 Jan 2013 | 2,180 | 951.8 | 3,381 |
| 02 Jan 2013 | 2,218 | 792.4 | 3,374 |
| 03 Jan 2013 | 2,333 | 1,008 | 3,688 |
| 04 Jan 2013 | 2,356 | 977.8 | 3,702 |
| 05 Jan 2013 | 2,250 | 794.7 | 3,660 |
| 06 Jan 2013 | 1,922 | 492.4 | 3,412 |
| 07 Jan 2013 | 2,036 | 585.5 | 3,620 |

- Next day (01 Jan 2013): about 2,180, likely between 952 and 3,381 (80% range).
- By 07 Jan 2013 the forecast is 2,036, 25% below the last complete day.
- Seasonal pattern: Friday is typically the high point (+169) and Sunday the low point (-266).
- Backtest on the last 7 days: the model was off by 1,063 on average (134%), better than simply using the same period last season (off by 2,020).
- The lower end of the range is cut off at 0, because the values can't be negative.

## Forecast: sum of cnt per month (exponential smoothing with level)

| Period | Forecast | Low (80%) | High (80%) |
|---|---|---|---|
| January 2013 | 123,700 | 93,890 | 152,900 |
| February 2013 | 123,700 | 79,530 | 160,300 |
| March 2013 | 123,700 | 71,430 | 171,600 |
| April 2013 | 123,700 | 63,040 | 181,200 |
| May 2013 | 123,700 | 53,720 | 190,500 |
| June 2013 | 123,700 | 48,390 | 198,600 |

- Next month (January 2013): about 123,713, likely between 93,894 and 152,856 (80% range).
- By June 2013 the forecast is 123,713, about the same as the last complete month.
- Backtest on the last 6 months: the model was off by 26,911 on average (19%), no better than simply using the last month repeated (off by 26,911).
- Seasonality needs at least two full years of months (26); this series has 24, so seasonal patterns are not modelled.

---

Statistics are computed from the data as loaded and cleaned in Data Boss. Associations and differences describe this data; they do not by themselves show cause.